

SECURE NETWORK FOR A SMALL RETAIL BUSINESS WITH REMOTE ACCESS

Group 9

Each of us played a vital role in ensuring the successful implementation of the small retail business network.

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1.0 Overview

This project aimed to design and configure a secure network infrastructure for a small retail business, ensuring internal segmentation, secure remote access, and robust management. The objective was to create a scalable and secure network that protects sensitive business data and provides controlled access to authorized personnel.

The scope of this project encompassed the design and implementation of the following key components:

- Network Design: Segmented network topology with VLANs for Sales, Inventory, and Admin departments
- Traffic Control: Access Control Lists (ACLs) for inter-departmental access and internet restrictions
- Remote Access: Secure VPN and SSH for remote management

- Management and Monitoring: SNMPv3, Syslog, and RADIUS for secure network management.

The project has been successfully implemented using Cisco Packet Tracer, which provides a secure, efficient, and scalable network infrastructure that meets the security and access requirements of the retail business.

2.0 Network Design

The network topology for the small retail business consists of three departmental VLANs (Sales, Inventory, and Admin) to segment traffic and ensure controlled communication between departments. It also include SNMPv3, Syslog, and RADIUS server for enhanced network management and monitoring.

VPN server is also included for secure remote access for authorized personnel.

2.1 VLAN Configuration:

ID	Name	IP Address	Assigned Ports
VLAN 10	Sales	192.168.10/24	Fa0/2 - Fa0/10
VLAN 20	Inventory	192.168.20/24	Fa0/11 - Fa0/20
VLAN 30	Admin	192.168.30/24	Fa0/21 - Fa0/24

2.2 Inter-VLAN Routing Configuration:

To enable inter-VLAN routing, sub-interfaces were created on the router for each VLAN. Each sub-interface is configured with encapsulation and an IP address.

VLAN	Sub Interface IP Address
VLAN 10	192.168.10.1
VLAN 20	192.168.20.1
VLAN 30	192.168.30.1

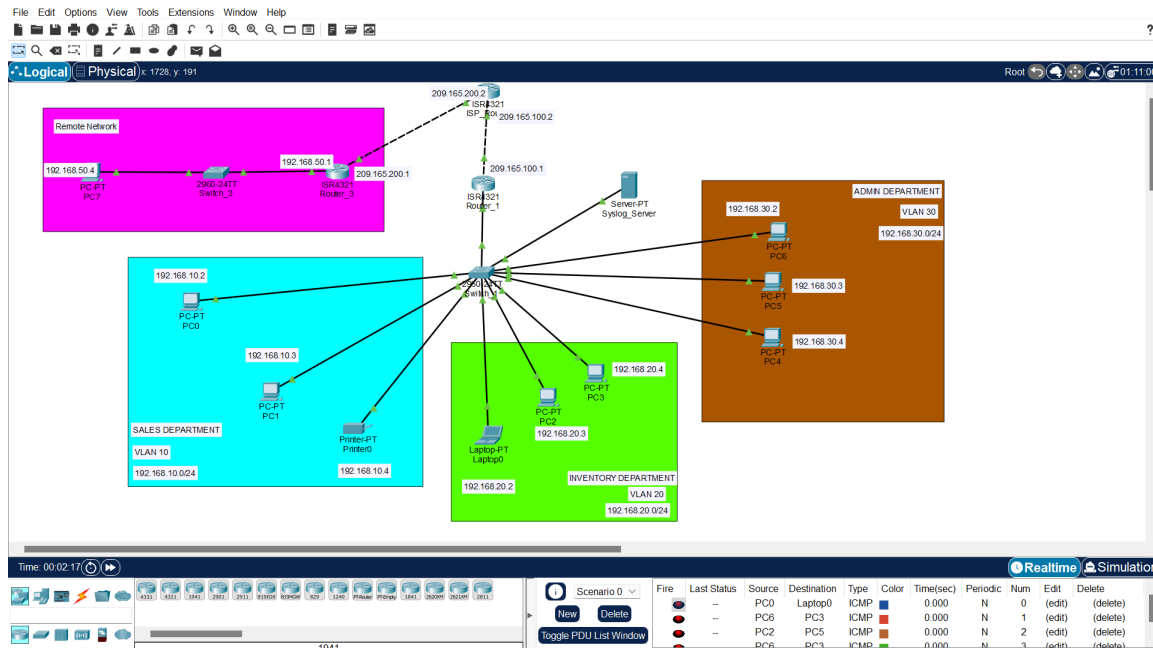
2.3 Network Devices:

Device Type	Device Name	IP Address	Purpose
2960-24TT	Switch_1	-	Segment the network and connect the end devices.
ISR4321	Router_1	209.165.100.1(Gig/0/0/1)	Routing and filtering traffic.
ISR4321	Router_2	209.165.100.2(Gig/0/0/0) 209.165.200.2(Gig/0/0/1)	Serve as ISP to provide internet connectivity.
ISR4321	Router_3	209.165.200.1(Gig/0/0/0) 192.168.50.1(Gig/0/0/1)	Enables secure remote access.
2960-24TT	Switch_3	-	Connects the remote end device.
Server-PT	Syslog_Server	192.168.10.6	Logging network events.

2.4 End Devices:

Device Type	Device Name	Network	IP Address
PC-PT	PC 0	VLAN 10	192.168.10.2
PC-PT	PC 1	VLAN 10	192.168.10.3
Printer-PT	Printer 0	VLAN 10	192.168.10.4
Laptop-PT	Laptop 0	VLAN 20	192.168.20.2
PC-PT	PC 2	VLAN 20	192.168.20.3
PC-PT	PC 3	VLAN 20	192.168.20.4
PC-PT	PC 4	VLAN 30	192.168.30.4
PC-PT	PC 5	VLAN 30	192.168.30.3
PC-PT	PC 6	VLAN 30	192.168.30.2
PC-PT	PC 7	Remote	192.168.50.4

2.5 Network Diagram:



3.0 Traffic Control with ACLs

To enhance network security, Access Control Lists (ACLs) were implemented. ACLs were configured to achieve two primary security objectives: minimizing the risk of malicious actors moving between departments and isolating sensitive data and resources.

This was implemented to:

- Restrict traffic between Sales, Inventory, and Admin departments.
- Allow necessary communication between departments and the entire network.
- Block unnecessary traffic.

Extended numbered ACLs (ACL 100-199) were utilized to filter traffic based on source and destination IP addresses, protocols, and ports.

4.0 Remote Access

To provide secure remote access to the network, a Virtual Private Network (VPN) and Secure Shell (SSH) protocol was configured. This ensures secure remote access and management, protecting sensitive business data and preventing unauthorized access to the network.

4.1 VPN Configuration:

The VPN protocol used is the Internet Protocol Security (IPSec) protocol. ESP (Encapsulating Security Payload) protocol is used to provide confidentiality, integrity, and authentication.

Authentication method include:

- Pre-Shared Key (PSK) authentication for secure tunnel establishment
- Username/Password authentication for remote users

The encryption Techniques used include:

- AES (Advanced Encryption Standard) with 256-bit keys for data encryption
- SHA(Secure Hash Algorithm)-HMAC(Hash-based Message Authentication Code) for data integrity and authentication.

4.2 SSH Configuration:

- The network devices is configured to use SSH version 2, which is more secure than version 1.
- password encryption is enabled to protect passwords from being displayed in plain text.
- Configured a local username and password for authentication.
- Restricted VTY access to SSH only

5.0 Management and Monitoring

To ensure robust network management and monitoring, the following protocols are implemented:

- SNMPv3 (Simple Network Management Protocol version 3) for Monitoring
- Syslog (System Logging) for Logging Network Events
- RADIUS (Remote Authentication Dial-In User Service) for Centralized Authentication

5.1 SNMPv3 configuration:

- Configured SNMPv3 on all network devices
- Implemented SNMPv3 views to restrict access to sensitive information
- Implemented Read only access to allow monitoring
- Implemented Read-write access to enable configuration changes
- Implemented Community string(password) to prevent unauthorized access

5.2 Syslog configuration:

- Configured Syslog servers to collect log data from network devices.
- Configured the router to send log messages to the Syslog server.
- Implemented debug-level logging to provide detailed information for troubleshooting.

5.3 RADIUS configuration:

- Configured RADIUS server to authenticate network device access
- Implemented RADIUS clients on network devices
- Configured RADIUS accounting to track user activity

6.0 Security Measures

To protect sensitive business data and provide controlled access to authorized personnel, the following security measures were implemented:

1. Segmented Network Topology: The implementation of VLANs (Sales, Inventory, Admin) isolates sensitive data and resources and reduce attack surface, limiting potential damage.

2. Access Control Lists (ACLs): The implementation of ACLs provides traffic filtering which prevents unauthorized access to the network. It also enforce the network segmentation by restricting access between departments.

3. Secure Remote Access:

- The implementation of IPsec VPN provides secure connectivity for remote workers. The encryption protects sensitive business data and the authentication ensures only authorized access.
- The implementation of SSH ensures secure authentication and encrypts all sessions, protecting sensitive data from eavesdropping and tampering.

4. Robust Management and Monitoring:

- SNMPv3 provides secure network monitoring and management and encrypts data to prevent unauthorized access.
- Syslog provides centralized logging of network events and enables log analysis and auditing.

- RADIUS Provides centralized authentication, authorization, and accounting (AAA) for network devices. Ensures secure access to network devices.